

Hey, everybody. Welcome to the Chapter 17 review. Today, we're going to focus on balance training concepts. If you've been tracking these in order, you know it's another expanded look at one of the components of integrated training. Technically, balance is defined as the ability to maintain one's center of gravity within their base of support. Maintaining postural control through balance is a fundamental component of performance, injury prevention, and rehabilitation after an injury.

Balance training is another one of those areas of fitness that may be viewed as non-traditional by some clients, so it'll be important for you to be able to justify and communicate the benefits of balance training exercises towards achieving their goals. After this chapter, you should be able to describe the scientific rationale for balance training, summarize its benefits for various client types with differing health goals, identify proper progression sequences, and employ methods used to effectively execute, instruct, and cue balanced training exercises.

Let's get into this in more detail. The center of gravity is a foundational key to the principle of balance. The center of gravity in the human body is located at the approximate midpoint, typically in the middle of the trunk. The base of support represents the area beneath the person that consists of every point of contact made between the body and the support they are on.

For instance, the width between the feet of a standing individual would define their base of support. If someone is standing still, the center of gravity can move within what we refer to as the limits of stability. The term "limit of stability" means the area within which a person can move their center of gravity without changing their base of support, like maintaining balance without moving their feet from the floor or falling.

Balance can be classified as static, semi-dynamic, or dynamic. This is based on positions being stationary or with limited or ever-changing stability. Static balance applies to situations when a person is trying to maintain postural control within a stationary position, like standing on one foot. Semi-dynamic balance is when a person is trying to maintain balance within a stationary limit of stability, but the base supporting the body is in movement. This would be similar to the movement you'd see in someone riding on a skateboard.

Lastly, dynamic balance is when a person works to maintain their center of mass over an ever changing limit of stability, like running on an uneven surface. Scientifically speaking, the balance mechanism involves three key sense-based systems. These include vision, vestibular, and somatosensation. Vision is typically used to provide information to the central nervous system about the body's location in space.

The vestibular senses, which are controlled by sensory receptors in the inner ear, provide the brain information about spatial orientation and the movement of the head in space. And somatosensation refers to the ability to feel changes in pressure on the skin, muscle length, and joint angles. This is especially important for balancing on unstable surfaces and for tasks that involve dynamic balance.

Balance training is a simple, often overlooked area of training that is going to be great for nearly everyone, and an area with lots of fun, creative ways to incorporate it into your programs. On one hand, you'll have serious athletes. And for them, research has shown athletes who perform balance training within an injury prevention program demonstrate improved performance and movement quality during landing tasks, greater muscle activity, and enhanced proprioception.

All of these things can reduce the risk of common lower extremity injuries, like ACL ruptures or ankle sprains. And these skills and improvements in function are going to carry through their entire season, and hopefully, athletic career, if it's something they incorporate into their ongoing training. On the flip side, balance training is also beneficial for older adults. Falls are one of the leading causes of injuries and health related complications in this population.

However, it's been shown those who regularly perform balance exercises have improved confidence and easier time maintaining body awareness and control, and reduced risk of taking those unwanted and often very harmful tumbles. With younger, older, or post-rehabilitation clients, you should implement progressive, systematic balance training in order to develop consistent long-term changes.

When you're designing a balance program, there are specific guidelines to follow, including proper exercise selection and progressive milestones. Many variables can also be manipulated when designing a balanced training program, including planes, ranges, speeds of motion, and the proprioceptive environment. For example, a single leg balance exercise could become more advanced over time by closing the eyes, passing a ball back and forth, or standing on an unstable surface.

You won't necessarily start with those variations, but manipulating one variable at a time, over time, can be a helpful way to ensure the exercise ramps up in difficulty and at the right pace. This will enable you to ensure your client has mastered one stage before moving on to the next. There are some common balance training progressions, like advancing from a stable to an unstable surface, or incorporating movement of the balancing leg through a full range of motion.

You can also have your clients combine hopping motions with a single-leg-stance landing. As with any training method, proper form and technique must be mastered to ensure proper muscle activation, and to avoid potential injury. Monitor the five kinetic chain checkpoints and emphasize quality over quantity whenever implementing this style of training.

There are a lot of different reasons why a person might have impaired balance. A musculoskeletal injury like a ligament sprain or a muscle pull, an alteration of sensory motor function due to something like a concussion, or even just the normal process of aging may contribute to deficits in balance. In all of these cases, the individual is going to benefit from balance training programs. Just remember, you will need to emphasize safety when you're designing a progressive balance training program, especially for clients with a current injury or history of injuries.

There are tons of creative exercise options and conditions you can use with clients in a way that's safe, progressive, and tailored to their needs. Training the balance system can be exciting and rewarding as you watch a client integrate strength, flexibility, and neuromuscular control into static and dynamic positions, and adapt to different and progressively more challenging variations.

Ultimately, you have the power through balance training to improve your client's health, wellness, and performance. Hopefully, all of this will get you started on the right foot. Now stay on that right foot, hold it for several seconds, and then hop around. Now close your eyes and do it again. See? Everyone can use some balance training, right?

I hope you're enjoying everything we've covered so far. More to come soon. Until then, be well and happy studying.